

José Manuel Requena Plens

Embedded Firmware & Software Engineer

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Summary

Embedded firmware and software engineer with 3+ years building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid **R&D** background. Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

Experience

R&D Software Developer

Apr 2023 – May 2026

Power Electronics — Solar + Storage

Llíria, Spain

- Reduced RTOS task stack usage by 30%–50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Improved embedded file-system stability, reducing access times under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated thousands of unit tests (Ceedling) and on-hardware integration tests (pytest): TCP/FTP performance and FTP security.
- Reviewed the team's merge requests, catching bugs and proposing improvements before integration.

Predoctoral Researcher (acoustics & ultrasound)

Jul 2021 – Apr 2023

UPV · CSIC · i3M / IIS La Fe

Valencia, Spain

- Designed acoustic metamaterials (metadiffusers and lenses) that reduce the footprint of conventional diffusers, validated with MATLAB and COMSOL simulations.
- Manufactured and characterized prototypes; published results at international conferences (Euronoise, Tecniacústica).
- Supervised a Master's thesis in Acoustic Engineering (UPV), from research design to defense.

Researcher in European Project

Feb 2020 – Jul 2021

UPV · European Space Agency (ESA) — IGIC

Gandía, Spain

- Designed and experimentally validated a noise-mitigation metamaterial for the VEGA rocket launch (ESA project AO/1-9479/18/NL/LvH), meeting 100% of project objectives with results accepted by ESA.
- Developed acoustic measurement software per ISO 10534-2 and ASTM 2611 ([A](#) | [Lab](#)) and a 3-axis robotic system for experimental measurements.
- Co-author of international publications/talks (Euronoise, ECSSMET).

Technical Skills

Firmware & embedded: C, STM32/ARM, ESP32/ESP-IDF, FreeRTOS, PlatformIO, Modbus RTU/TCP, I2C/SPI/UART, JTAG/SWD, Secure Boot, Doxygen, instrumentation (oscilloscope, signal generator, logic analyzer)

Software & languages: Python, Go, C++, Bash, TypeScript, Git, Docker, CMake, Clang tooling

Quality, CI/CD & DevSecOps: GitHub Actions, GitLab CI, Jenkins, SonarQube, GoReleaser, Ceedling, pytest, Playwright, static analysis/SAST (CodeQL, gitleaks), pre-commit

Networking: TCP/IP, MikroTik RouterOS, WireGuard, IPv6, QUIC/HTTP3, DNS

Security & cryptography: TLS/mTLS, PKI (public-key infrastructure), applied cryptography (AEAD, key derivation/KDF), hardening, CrowdSec

AI & agentic: Model Context Protocol (MCP), AI coding agents (Claude Code, Copilot), LLM integration

Simulation & measurement: MATLAB, COMSOL Multiphysics, LabVIEW, acoustic instrumentation (laser & vibration accelerometers, microphones)

Open Source Projects

gitlab-mcp-server — Go · MCP server (Model Context Protocol) 14 ★ • ~23k downloads • Glama / mcp.so / Cursor

GitLab MCP server exposing 1,000+ operations to AI assistants through a dynamic 2-tool design (*find/execute*), with stdio/HTTP/OAuth transports.

PyOctaveBand — Python · signal processing 96 ★ • PyPI

Octave-band and fractional octave-band filtering library for time-domain signals.

cs-routeros-bouncer — Go · network security 10 ★

CrowdSec bouncer for MikroTik RouterOS; manages address lists and firewall rules via the RouterOS API.

Cloudflare-DNS-Updater — Bash · DevOps 26 ★

Dynamic DNS client for Cloudflare (IPv4/IPv6) with a bats test suite and coverage.

Education

PhD Programme in Health and Wellness Technologies 2023

Polytechnic University of Valencia (i3M / UMIL) Valencia, Spain

Research on acoustic metamaterials for medical imaging and ultrasound beam control (programme not completed; redirected to industrial software development).

Master's Degree in Acoustic Engineering 2019

Polytechnic University of Valencia (EPSG) Gandía, Spain

Graduated with honors; Master's thesis awarded a special distinction.

Degree in Telecommunication Engineering (Sound & Image) 2018

University of Alicante (EPS) Alicante, Spain

Two specializations: Acoustic Engineering and Audiovisual Technology.

Vocational Higher Degree in Sound & Technical Degree in Electronic Systems 2009 – 2011

IES Salesianos · IES Luis García Berlanga Elche / Alicante, Spain